A Sequential Decision-Making Framework for a Two-Player Simplified Real Time Strategy Game

ASEN 5264 — Decision Making Under Uncertainty

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I. Introduction

This project develops a two-player strategy game in which each player manages workers, marines, and resources with the goal of eliminating all enemy marines. The project is motivated by the fields of deterrence theory and conflict simulation where actors need to make decisions based on available information in order to achieve strategic goals. The project creates a simplified version of real time strategy games which have been studied extensively in reinforcement learning, most famously AlphaStar. The game is formalized as a Markov Decision Process and solved using methods from the course, with increasing model complexity across three tiers of completion. The first tier explores solution of the fully observable MDP. First, value iteration is used to solve for an optimal policy against a deterministic policy of the adversary and Monte Carlo Tree Search (MCTS) to evaluate online planning effectiveness. Reinforcement learning is also applied, specifically Q-learning and Deep Q-Network (DQN) learning to learn against this optimal policy. The second tier extends the problem formulation in two branches. First, it is extended to a Partially Observable Markov Decision Process (POMDP), where the player does not have full state knowledge, which is approximately solved with both the Q-Value MDP approximation (QMDP) and Partially Observable Monte Carlo Planning (POMCP). Second, it is simulated as a Markov Game (MG), where both players have dynamic gameplay policies. The Markov Game is solved using fictitious self-play, in which each player iteratively best-responds against the running average of all past opponent policies, guaranteeing convergence to a Nash equilibrium in this zero-sum setting. The third tier would combine the MG and POMDP into a Partially Observable Stochastic Game (POSG). Among MDP solvers, value iteration achieved the highest win rate but at the cost of full state-space enumeration; DQN approached comparable performance with significantly better scalability. Under partial observability, QMDP proved robust to limited opponent state information but brittle to opponent policy mismatch, where POMCP's online planning maintained stronger performance by avoiding reliance on a precomputed value function. Under the Markov Game formulation, fictitious self-play converged to a near-even Nash equilibrium after 100 iterations; when MDP solvers were evaluated against the resulting FSP opponent policy, DQN outperformed value iteration (73% vs. 60%), while Q-learning degraded significantly, highlighting the sensitivity of offline planners to opponent policy assumptions. Time restrictions and problem complexity did not allow for continuing into the third tier and modeling a POSG.

II. Background and Related Work

The main motivation behind this project was to simulate deterrence as a partially observable stochastic game (POSG). Deterrence is typically modeled using game theory between multiple agents but has not been framed as a Markov Decision Process (MDP). However, modeling deterrence as an MDP needed to be dramatically simplified to be completed within the time allotted. Adjacent to the concept of deterrence is conflict simulation. Real-time strategy (RTS) games are the most analogous framework to conflict simulation and are a platform that can be integrated with reinforcement learning frameworks. RTS games possess many attributes that make reinforcement learning difficult: partial observability, large state and action spaces, rapid response times, and evolving strategies [1]. In the field of reinforcement learning, AlphaStar is considered one of the most significant achievements [2], where it was trained to play StarCraft II (SC2). AlphaStar first used supervised learning on 971,000 previously recorded human matches to produce a diverse initial policy. AlphaStar then used V-trace, an off-policy actor-critic algorithm, as its primary reinforcement learning update. On the multi-agent side, AlphaStar used a league training procedure based on prioritized fictitious self-play (PFSP) to compute best responses against a diverse population of opponents [2]. Other RTS games on which reinforcement learning has been applied include StarCraft: Brood War [3] and Warcraft III [4]. Additionally, due to the complexity of full commercial RTS games, lightweight environments have been developed specifically for reinforcement learning research, including microRTS [5], Mini-RTS [6], and Deep RTS [7]. Each environment has its own trade-offs in complexity and simulation speed, but all were designed as platforms to test reinforcement learning algorithms more efficiently. This project, due to time constraints, simplifies these ideas further. The state and action spaces were reduced significantly, but various algorithms could still be applied to find policies capable of reaching terminal conditions. This

project formulation was designed to capture the core dynamics of RTS games at their simplest level — workers, marines, and resources — while incorporating reinforcement learning and game-theoretic concepts to solve for optimal policies.

III. Problem Formulation

A. State Space

The state for each player $i \in \{1, 2\}$ is (w_i, m_i, r_i), representing workers, marines, and resources. The joint state is $s = (w_1, m_1, r_1, w_2, m_2, r_2)$, plus a terminal state s_{term}. Both players start at $w = m = r = 1$. Resources increase by w_i each turn. Workers, marines, and resources for each player are limited to $w_{\max} = m_{\max} = r_{\max} = 10$, yielding approximately 3.5M total states.

B. Action Space

Each player selects from the following action set each turn:

$$\mathcal{A} = \{\texttt{train worker}, \texttt{train marine}, \texttt{attack}\}$$

C. Transition Model

Table I summarizes the transition model. If either player has $m_i = 0$, the game transitions to s_{term} with probability 1 regardless of action. The probability, $P(k_A, k_B)$, resulting from an `attack` action is given by the binomial distribution

$$P(k_A, k_B) = \binom{m_B}{k_A} p^{k_A} (1-p)^{(m_B - k_A)} \times \tag{1}$$

$$\binom{m_A}{k_B} p^{k_B} (1-p)^{(m_A - k_B)} \tag{2}$$

where k_A, k_B represent the marines loses during an attack engagement and the propability of success was set to $p = \frac{1}{2}$.

D. Reward

$R = 1$ upon reaching s_{term} as the winner and $R = -1$ for the defeated player. This basic reward structure makes the MDP a zero-sum game from a game theory perspective.

IV. Solution Approach

A. Tier 1: Markov Decision Process Solutions

1) Value Iteration on the Fully Observable MDP: The game is formalized as a fully observable MDP with pre-computed transition matrices T_a for each Player 1 action a. Player 2's default action policy, `alternating_training_attack`, alternates between training marines and workers unless both are saturated ($m_2 = w_2 = 10$), in which case the attack action is selected. This MDP is solved via value iteration (Algorithm 1) to obtain an optimal policy π^*, iterating the Bellman update

$$U'(s) = \max_{a \in \mathcal{A}} \left(R(s,a) + \gamma \sum_{s' \in \mathcal{S}} T(s' \mid s, a)\, U(s') \right) \tag{3}$$

until $\|U - U'\|_\infty < \varepsilon$. The greedy policy is then extracted as $\pi^*(s) = \arg\max_a \big(R(s,a) + \gamma\, \mathbb{E}[U^*(s')]\big)$.

Algorithm 1 Value Iteration [8]

Given: MDP $(\mathcal{S}, \mathcal{A}, R, T, \gamma)$, tolerance ε
Initialize U, U' (differently)
repeat
  $U \leftarrow U'$
  for each $s \in \mathcal{S}$ **do**
    $U'(s) \leftarrow \max_{a \in \mathcal{A}} \left(R(s,a) + \gamma \sum_{s' \in \mathcal{S}} T(s' \mid s, a)\, U(s') \right)$
  end for
until $\|U - U'\|_\infty \le \varepsilon$
return U'

2) Online Planning with MCTS: Since value iteration requires full-state enumeration, Monte Carlo Tree Search (MCTS/UCT, Algorithm 2) is also implemented as an online planner. From each state encountered during gameplay, MCTS runs $N = 1000$ simulations, each of which selects actions via the UCB criterion $Q(s,a) + c\sqrt{\ln N(s)/N(s,a)}$ with $c = \sqrt{2}$, expands the tree at leaf nodes, estimates value via a random rollout to depth 50, and backs up the discounted return. The agent then acts greedily with respect to the resulting Q estimates.

Algorithm 2 Monte Carlo Tree Search (MCTS/UCT) [8]

function MCTS(s)
  for $k = 1, \ldots, m$ **do**
    SIMULATE(s)
  end for
  return $\arg\max_{a \in \mathcal{A}} Q[(s,a)]$
end function

function SIMULATE(s, d)
  if $d \le 0$ **then**
    return $U(s)$ ▷ value estimate / rollout
  end if
  if s not yet in tree **then**
    for each $a \in \mathcal{A}$ **do**
      $N[(s,a)] \leftarrow 0; \quad Q[(s,a)] \leftarrow 0$
    end for
    return $U(s)$
  end if
  $a \leftarrow \arg\max_a \left[Q[(s,a)] + c\sqrt{\dfrac{\ln N(s)}{N(s,a)}} \right]$
  $s', r \leftarrow \mathrm{TR}(s,a)$
  $q \leftarrow r + \gamma \cdot$ SIMULATE(s', $d-1$)
  $N[(s,a)] \leftarrow N[(s,a)] + 1$
  $Q[(s,a)] \leftarrow Q[(s,a)] + \dfrac{q - Q[(s,a)]}{N[(s,a)]}$
  return q
end function

3) Reinforcement Learning: Q-Learning and DQN: Two reinforcement learning methods are applied. Q-learning (Algorithm 3) maintains a tabular estimate $Q(s,a)$ updated via the TD error at each step, with $\alpha = 0.1$, $\gamma = 0.95$, and ε

TABLE I
TRANSITION MODEL

Action	Condition	Next State	Prob.
train worker	$r \geq 1$	$w' = w + 1,\ r' = r - 1$	0.9
train marine	$r \geq 1$	$m' = m + r,\ r' = 0$	0.9
attack	always	$m'_A = m_A - k_A,\ m'_B = m_B - k_B$	$P(k_A, k_B)$
terminal	$m_i = 0$	s_{term}	1.0

decaying linearly from 0.2 to 0.05.

Algorithm 3 Q-Learning [8]

$Q(s,a) \leftarrow 0$ for all (s,a); $s \leftarrow s_0$
loop
  $a \leftarrow \begin{cases} \arg\max_a Q(s,a) & \text{w.p. } 1-\varepsilon \\ \text{rand}(\mathcal{A}) & \text{otherwise} \end{cases}$
  $r \leftarrow \texttt{act!}(env, a)$; $s' \leftarrow \texttt{observe}(env)$
  $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$
  $s \leftarrow s'$
end loop

DQN (Algorithm 4) replaces the Q-table with a neural network $Q_\theta(s,a)$, enabling generalization to larger state spaces. The network maps the 7-dimensional state to a 256-unit hidden layer (ReLU) followed by a 3-element action output. Training uses a circular replay buffer (capacity 10^5), mini-batches of 50, target-network updates every 50 steps, and Adam with learning rate 5×10^{-4}. Exploration decays exponentially from $\varepsilon_0 = 0.5$ to $\varepsilon_{\min} = 0.01$.

Algorithm 4 Deep Q-Network (DQN)

Initialize Q_θ; target network $Q_{\theta'} \leftarrow Q_\theta$; replay buffer $\mathcal{D} \leftarrow \emptyset$
loop
  $a \leftarrow \begin{cases} \arg\max_a Q_\theta(s,a) & \text{w.p. } 1-\varepsilon \\ \text{rand}(\mathcal{A}) & \text{otherwise} \end{cases}$
  $r \leftarrow \texttt{act!}(env, a)$; $s' \leftarrow \texttt{observe}(env)$; store (s,a,r,s') in $\mathcal{D}$
  Sample mini-batch $\mathcal{B}$ from $\mathcal{D}$
  $\theta \leftarrow \theta - \alpha \nabla_\theta \frac{1}{|\mathcal{B}|} \sum_{(s_j,a_j,r_j,s'_j)\in\mathcal{B}} \big(r_j + \gamma \max_{a'} Q_{\theta'}(s'_j, a') - Q_\theta(s_j, a_j)\big)^2$
  Periodically: $\theta' \leftarrow \theta$ ▷ sync target network
  $s \leftarrow s'$
end loop

B. Tier 2: POMDP and Markov Game Solutions

1) QMDP: The game model is extended to a POMDP where Player 1 observes its own personnel and resources exactly but receives only a discretized observation of Player 2's personnel with granularity n. QMDP approximates the POMDP value function by assuming full observability after one step, yielding

$$Q_{\text{QMDP}}(b,a) = b^\top (R + \gamma T_a V^*) \tag{4}$$

$$\pi_{\text{QMDP}}(b) = \arg\max_a Q_{\text{QMDP}}(b,a). \tag{5}$$

2) POMCP: While QMDP is simple to implement, it assumes next-step full observability and requires a full-state value function. POMCP (Algorithm 5) instead combines MCTS with a particle-based belief, indexing the search tree by action–observation history h rather than state. Each call draws 200 particles from the current belief, runs 5000 simulations using UCB, and performs random rollouts to depth 20 from leaf nodes.

Algorithm 5 POMCP [8]

function POMCP(b)
  for $k = 1, \ldots, m$ **do**
    $s \sim b$ ▷ sample state from belief
    SIMULATE(s, h_0, d)
  end for
  return $\arg\max_{a\in\mathcal{A}} Q[(h_0, a)]$
end function

function SIMULATE(s, h, d)
  if $d = 0$ **then**
    return 0
  end if
  if $h \notin$ tree **then**
    for each $a \in \mathcal{A}$ **do**
      $N[(h,a)] \leftarrow 0$; $Q[(h,a)] \leftarrow 0$
    end for
    return ROLLOUT(s, d) ▷ expansion + rollout
  end if
  $a \leftarrow \arg\max_a \left[Q[(h,a)] + c\sqrt{\frac{\ln N(h)}{N(h,a)}} \right]$
  $s', o, r \leftarrow G(s,a)$ ▷ generative model sample
  $q \leftarrow r + \gamma \cdot$ SIMULATE(s', $h \cdot ao$, $d-1$)
  $N[(h,a)] \leftarrow N[(h,a)] + 1$
  $Q[(h,a)] \leftarrow Q[(h,a)] + \frac{q - Q[(h,a)]}{N[(h,a)]}$
  return q
end function

3) Fictitious Self-Play: All previous approaches fixed the opponent's policy, reducing the game to a single-agent MDP. A Markov game extends the MDP framework to the multi-agent setting by defining a joint action space and per-player reward functions. Two solution methods are applied in unison:

alternating best response (Algorithm 6) and fictitious self-play (Algorithm 7).

In ABR each player in turn freezes the opponent's current policy, reduces the MG to a single-agent MDP, and solves for a best response via value iteration. This can oscillate without converging since each player reacts only to the opponent's most recent policy. Fictitious self-play corrects this by maintaining a running average of each player's past best responses as a mixed strategy $\sigma^i(a \mid s)$, represented via action-count arrays $n^i[s, a]$ incremented after each solve. Each player best-responds against the opponent's empirical average σ^{-i}. In finite two-player zero-sum games, fictitious play is guaranteed to converge to a Nash equilibrium.

The combined transition and reward for Player 1 are marginalized over Player 2's mixed strategy:

$$T'(s' \mid s, a^1) = \sum_{a^2} T(s' \mid s, a^1, a^2)\, \sigma^2(a^2 \mid s), \tag{6}$$

$$R'(s, a^1) = \sum_{a^2} R(s, a^1, a^2)\, \sigma^2(a^2 \mid s), \tag{7}$$

and Player 1 runs value iteration against this marginalized MDP. The same procedure applies symmetrically for Player 2 using an inverted-state representation.

Algorithm 6 Alternating Best Response [8]

Initialize policies π_0^1, π_0^2 arbitrarily
for iteration $k = 1, 2, \ldots, K$ **do**
  Fix π_{k-1}^2; reduce MG to single-agent MDP for P1
  $\pi_k^1 \leftarrow \mathrm{VI}\big(\mathrm{env}_{P1}(\pi_{k-1}^2)\big)$ ▷ P1 best response
  Fix π_k^1; reduce MG to single-agent MDP for P2
  $\pi_k^2 \leftarrow \mathrm{VI}\big(\mathrm{env}_{P2}(\pi_k^1)\big)$ ▷ P2 best response
end for
return $\pi_K^1,\ \pi_K^2$

Algorithm 7 Fictitious Self-Play [8]

Initialize $n^1[s, a] \leftarrow 0$; seed n^2 from initial P2 policy
for iteration $k = 1, 2, \ldots, K$ **do**
  $\sigma^2 \leftarrow \mathrm{normalize}(n^2)$ ▷ P2 empirical average policy
  $\pi_k^1 \leftarrow \mathrm{VI}\big(\mathrm{env}_{P1}(\sigma^2)\big)$ ▷ P1 best response to σ^2
  $n^1[s, \pi_k^1(s)] \mathrel{+}= 1$ for all $s \in \mathcal{S}$
  $\sigma^1 \leftarrow \mathrm{normalize}(n^1)$ ▷ P1 empirical average policy
  $\pi_k^2 \leftarrow \mathrm{VI}\big(\mathrm{env}_{P2}(\sigma^1)\big)$ ▷ P2 best response to σ^1
  $n^2[s, \pi_k^2(s)] \mathrel{+}= 1$ for all $s \in \mathcal{S}$
end for
return $\sigma^1 \leftarrow \mathrm{normalize}(n^1)$, $\sigma^2 \leftarrow \mathrm{normalize}(n^2)$

C. Tier 3: Partially Observable Stochastic Game

Tier three would combine the concepts of POMDPs and MGs into a single framework. However, due to time constraints and the complexities of POSGs, tier three experimentation was not able to be acheived.

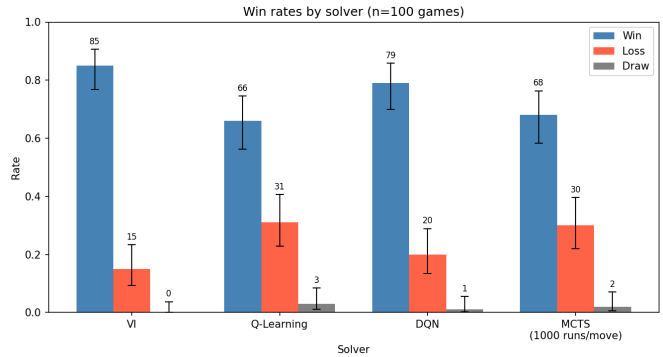

Fig. 1. MDP Solver Performance vs. `alternating_training_attack`. Error bars show 95% Wilson confidence intervals.

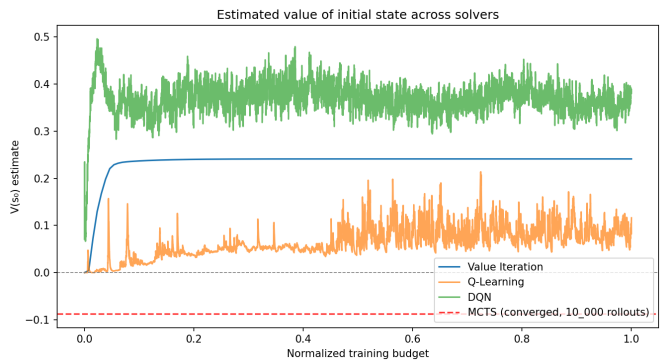

Fig. 2. Initial-state Value Estimate vs. normalized training/planning effort for each MDP solver

V. Results

A. Tier 1: MDP Solver Performance

All MDP solvers were evaluated from the Player 1 perspective against a fixed Player 2 policy using the following parameters to define the game environment and solver setup:

- Initial State: Workers = Marines = Resources = 1
- Player 2 policy: `alternating_training_attack`
- Number of episodes: 10,000
- Number of games to evaluate win rate: 100
- Discount factor: 0.95

As expected, value iteration achieved the best win rate (58%), since it converges to the optimal value function, from which a greedy policy can be directly extracted. DQN achieved a 50% win rate, and Q-learning and MCTS trailed further behind (see Figure 1). However, this performance comes at the cost of additional computation time, since the value iteration solution method iterates over the entire state space, where approximate solvers need only a sufficient sample of states. For the current state dimensions, this is very tractable, converging in approximately 30 seconds, but future increases in complexity - higher troop or resource limits, more types of personnel, more action options, etc. - can increase the state

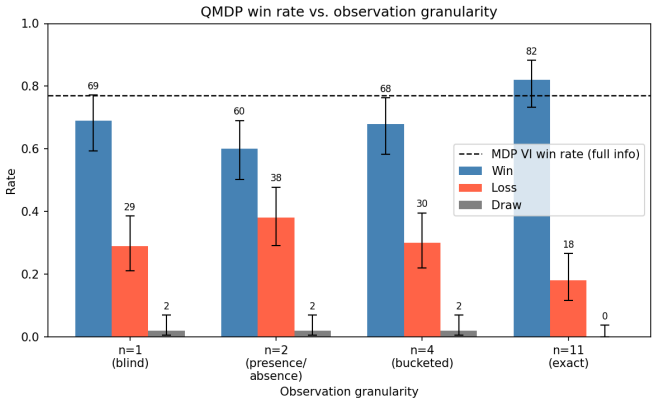

Fig. 3. QMDP Win Rate vs. Observation Granularity (n). Error bars show 95% Wilson confidence intervals.

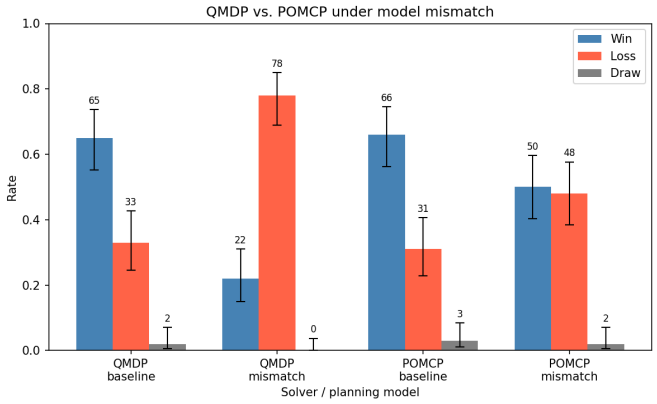

Fig. 4. Performance Under P2 Model Mismatch (planner assumes uniform-random P2; simulation uses `alternating_training_attack`). Error bars show 95% confidence intervals.

space size exponentially, and value iteration would quickly become intractable as it scales with $|\mathcal{S}|^2$. Q-learning scales better than value iteration ($|\mathcal{S}|$), but this could still become intractable, and as seen in Figure 1 Q-learning shows the worst performance of the implemented solvers. DQN and MCTS both attempt to minimize this curse of dimensionality: MCTS by exploring only a tree of reachable states, which would scale well to large state spaces, though it trades quality for this scalability, and may require significantly increased run counts in larger reachable state spaces to achieve acceptable performance. Since DQN approximates the value function with a neural network, it can generalize to unvisited states, and for a well-behaved value function can understand a very large state space without much added computational cost. This scalability appears to come at a much lower cost than MCTS for this problem, as DQN performs nearly as well as value iteration, making it a prime contender for use in future problem evolutions.

In Figure 2, the estimation errors from the other three solvers can be seen more directly. Q-learning and MCTS both underestimate the value of the initial state provided by value iteration, MCTS quite significantly, and DQN overestimates it. In future work, any chosen solver - likely DQN, based on its current performance - could be modified to account for some of this error, e.g., by implementing double Q-learning to remove maximization bias.

B. Tier 2: POMDP Observability

To explore the impacts of limited observation of the adversary's state, the QMDP solver was used in scenarios with varying levels of observability (no observability, presence or absence of personnel, bucketed estimate of personnel, and full observability). Each scenario was evaluated using the following parameters:

- Player 2 policy: `alternating_training_attack`
- Number of games: 100
- Step limit: 500

As expected, QMDP achieved the best win rate in the exact observation scenario, decreasing the observability of the opponent's state only decreased performance by a small margin. The performance for each of the bucketed, presence/absence, and blind scenarios is very similar, with the confidence intervals mostly overlapping. This indicates that knowledge of the adversary's population status is minimally important for winning the game. Stated another way, the best policy for Player 1 does not vary significantly based on Player 2's state, and QMDP's single-step lookahead is sufficiently capable for this POMDP formulation, achieving close to the MDP win rate even when observability is limited.

C. Tier 2: Model Mismatch and POMCP

In order to determine the effect of other sources of uncertainty, an additional scenario was constructed where the solver has no knowledge of Player 2's policy: uncertainty in this scenario comes not just from a partially observable state, but also from ambiguity over what actions the opponent will play from any given state. This was effectuated for the QMDP solver by building a value function from a uniform random Player 2 policy, but allowing Player 2 to instead play the `alternating_training_attack` policy in subsequent gameplay. Following an observed drop in QMDP's effectiveness under this uncertainty, the POMCP solver was also implemented to compare performance under the same conditions. The solver and gameplay parameters from the observation granularity investigation were reused, but here the enemy state remained fully unobservable (`n_obs_levels=1`).

The introduction of policy uncertainty significantly decreases QMDP's win rate, indicating that this POMDP solution method is brittle to an inaccurate value function - unsurprisingly for an offline planner. POMCP's results, as an online planner, are markedly different: it slightly outperforms QMDP in the baseline case (within the margin of error), but takes significantly longer to run - approximately 3x QMDP's runtime - as its Monte Carlo-based estimate of the value of a

state must be built at each step. POMCP, however, can build a reasonably accurate value estimate regardless of the prior knowledge QMDP relies on, and its higher performance in the "mismatch" scenario demonstrates this robustness.

D. Tier 2: Markov Game and Fictitious Self-Play

The previous approaches did not have multi-agent interaction. Formulating the problem as a Markov Game, fictitious self-play was impelemented to have actual agent interaction to improve their respective policies. For a simulated fictitious self-play, 100 iterations of alternating best response using value iteration were used to converge to a best response policy of each player. Since the reward structure makes this a zero-sum game, the initial value function for both players should converge to zero resulting in a mixed Nash equalibrium. Figure 5 shows that both player's value of the initial state for the average policy converged to zero over 100 iterations as expected. Since the values converged to zero, the mixed Nash equalibrium policies were approximated. The average policy entropy at the initial state s_0 is tracked across iterations to measure convergence of the mixed strategy. For a mixed policy σ^t over the action set $\mathcal{A}$, entropy is defined as

$$H(\sigma^t) = -\sum_{a \in \mathcal{A}} \sigma^t(a) \log \sigma^t(a) \tag{8}$$

where $\sigma^t(a)$ is the probability assigned to action a at FSP iteration t. Figure 6 shows that entropy for each policy converging around .1 nats. With a maximum possible entropy value of 1.099 nats (uniform distribution), a convergence value of .1 nats indicates that each players' policy is almost deterministic at each state. After the 100 iterations of alternating best response, figure 7 shows the win rates for both players after 1,000 simulated games. As expected, each player won approximately 50% of the time.

To measure how the MDP solvers perform against an improved opponent policy, player 2's FSP-derived mixed policy was frozen and used in place of the deterministic `alternating_training_attack` policy. Figure 8 shows the resulting win rates. Against the FSP opponent, DQN achieves the highest win rate (73%), surpassing value iteration (60%), a reversal of their relative ordering under the deterministic policy. This suggests that DQN's generalization through function approximation is better suited to opponents whose behavior is less predictable, while value iteration's exact solution is calibrated specifically to the policy used during planning. Q-learning degrades most severely under the FSP opponent (21% win rate), consistent with its poor sample efficiency and sensitivity to opponent policy during training. MCTS maintains moderate performance (56%), benefiting from its online planning which does not depend on a precomputed value function.

Figure 9 shows the estimated value of the initial state across solvers when trained against the FSP opponent. Value iteration converges near zero, consistent with the near-even game outcome expected against a Nash equilibrium policy. DQN's value estimate is substantially higher (≈ 0.3), a larger overestimation bias than observed against the deterministic opponent, suggesting that DQN's neural network exploits statistical regularities in the FSP opponent that do not generalize to held-out games. Q-learning produces a slightly negative estimate, reflecting its inability to find a competitive policy against the adaptive opponent.

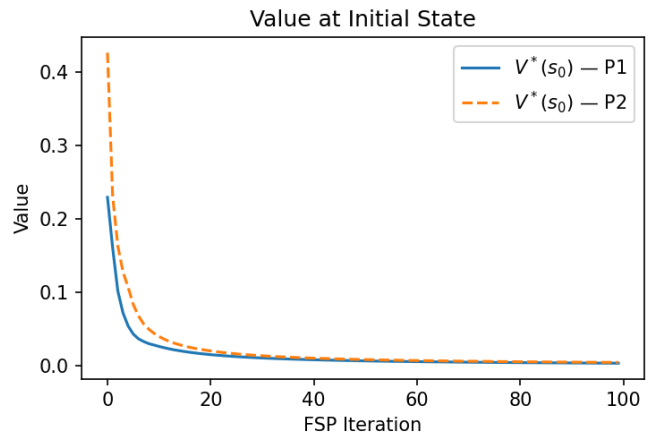

Fig. 5. Value at Initial State s_0 for each player over 100 FSP iterations. Both values converge toward zero, consistent with a mixed Nash equilibrium in this zero-sum game.

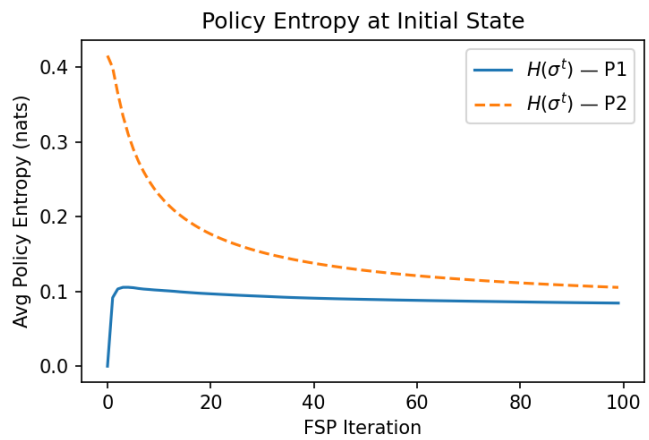

Fig. 6. Average policy entropy $H(\sigma^t)$ at s_0 for each player over 100 FSP iterations. Entropy decreases as the average policy concentrates on a dominant action, indicating convergence toward a near-deterministic Nash equilibrium strategy.

VI. Conclusion

This work formalized a two-player resource-and-combat game as an MDP and evaluated four solvers under full and partial observability. Value iteration produced the highest win rate (58%) against the fixed opponent policy, while DQN reached 50%—an 8-point gap, but achieved with substantially better scalability to larger state spaces, making it the most

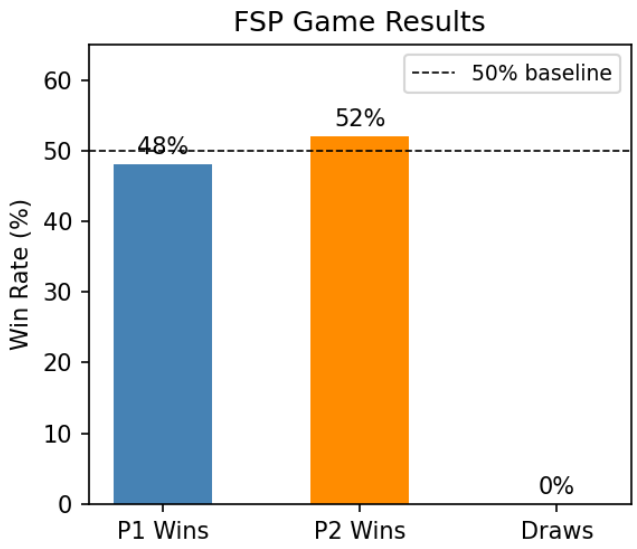

Fig. 7. Win rates over 1000 games using mixed Nash equilibrium policies from FSP. Results are approximately even (48% P1, 52% P2), consistent with convergence to a mixed Nash equilibrium in this zero-sum game.

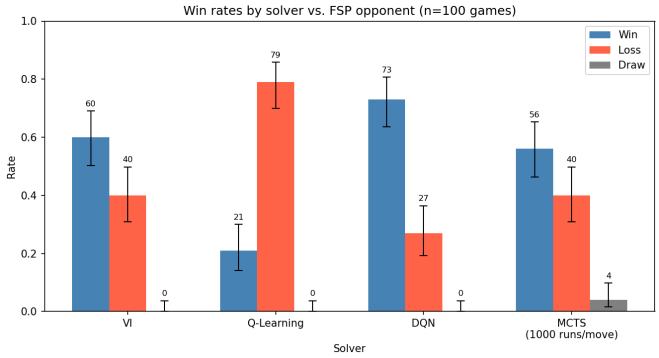

Fig. 8. Win rates for each MDP solver against the FSP-trained P2 opponent (100 games). VI and DQN achieve the highest win rates (60% and 73%), while Q-Learning struggles against the adaptive opponent policy.

practical choice for future problem extensions. Q-learning and MCTS both underperformed relative to their computational cost at this problem scale.

Under partial observability, QMDP demonstrated that knowledge of the opponent's unit counts has limited impact on win rate: performance degraded only modestly as observability decreased from perfect to fully blind. The more significant challenge proved to be opponent policy uncertainty: QMDP's win rate dropped from 40% to 15% under policy mismatch, while POMCP's online planning reduced this sensitivity.

Extending to a Markov Game via fictitious self-play produced a near-even Nash equilibrium (48%/52% win split over 1000 games), confirming convergence in this zero-sum setting. Evaluating the MDP solvers against the FSP-derived opponent

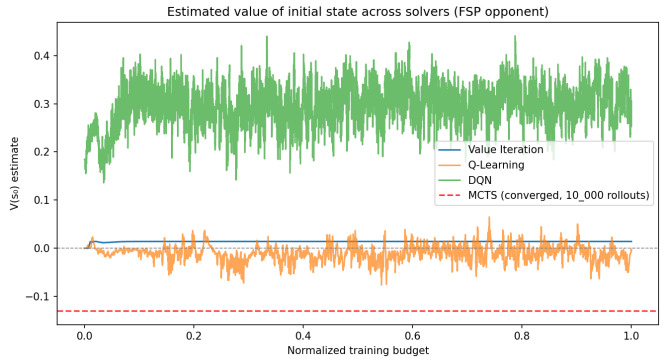

Fig. 9. Estimated value of the initial state $V(s_0)$ across solvers against the FSP opponent. DQN overestimates value due to function approximation bias; VI converges near zero, consistent with the near-even game outcome.

revealed that DQN's generalization advantage becomes most pronounced against adaptive opponents, surpassing value iteration's win rate despite value iteration's theoretical optimality against fixed policies.

Future work could increase state space complexity (higher troop and resource limits, additional unit types) to stress-test DQN's scalability advantage and further motivate POMCP over offline planners. Within the current formulation, double Q-learning could address the overestimation bias observed in DQN's value estimates. Lastly, the POSG framework can be explored to see how multi-agent interaction develops under partial observability.

Contributions

Jeremiah and Michael jointly developed the initial idea for a two-player, simplified, RTS-style game. Jeremiah implemented the initial MDP environment, value iteration, Q-learning, MCTS, and POMDP solvers. Michael converted the initial MDP and solvers into an object-oriented framework for ease extension and integration. He maintained version control of the code base via GitHub. Michael implemented DQN and the Markov Game framework. Both contributed to their respective sections in the final report and overall proofreading.

All algorithms presented in this report, value iteration, Q-learning, MCTS, QMDP, POMCP, alternating best response, and fictitious self-play, were implemented from scratch without relying on off-the-shelf reinforcement learning libraries such as Stable-Baselines or RLlib. DQN was implemented from scratch but used PyTorch for its neural network layers and optimizer. Otherwise, basic python libraries were used for non-reinforcement learning purposes.

The authors grant permission for this report to be posted publicly.

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